

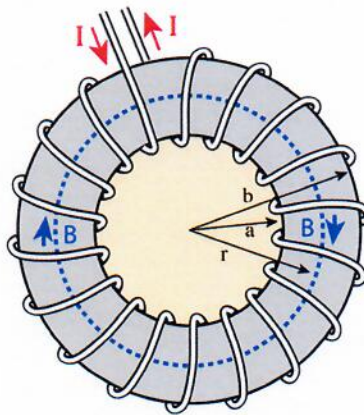
ELCT 361: Electromagnetics -Tutorial Problems

Chapters 6-1, 6-2, 7-1, and 7-2

Student Name:

Student ID or Matric Number:

Q1. A closely wound toroidal coil with an air core having N turns and carrying current I is shown below. (μ_0 : magnetic permeability of air core material)



- (a) Determine magnetic flux density, $\vec{B}$, circulating in the core based on "Ampere's Law".
- (b) Find the *mmf*.
- (c) Find magnetic flux, Φ inside the air core.
- (d) Find the *inductance* and *magnetic stored energy*
- (e) Find the *magnetic reluctance* of solenoid and draw an *equivalent magnetic circuit*.

Based on Ampere's Law, μ_0 , N -turns, I

$$(a) \int_S (\nabla \times \vec{B}) \cdot d\vec{s} = \oint_C \vec{B} \cdot d\vec{l} = N \int_S \mu_0 \vec{J} = \mu_0 I N, \quad \vec{B} = B_\phi \hat{a}_\phi \quad dl = r d\phi \hat{a}_\phi$$

$$\int B_\phi \hat{a}_\phi \cdot \hat{a}_\phi r d\phi = B_\phi \int_0^{2\pi} r d\phi = B_\phi 2\pi r = N \mu_0 I$$

$$\vec{B} = \hat{a}_\phi \frac{N \mu_0 I}{2\pi r}$$

$$(b) \text{ mmf} = N \mu_0 I$$

$$(c) \quad \vec{\Phi} = \int_S \vec{B} \cdot d\vec{S} = BS \quad (a < r < b) \quad S = \pi r^2 = \pi \left(\frac{b-a}{2}\right)^2$$

$$\vec{\Phi} = BS = \frac{N\mu_0 I}{2\pi r} \cdot \pi \left(\frac{b-a}{2}\right)^2 = \frac{N\mu_0 I (b-a)^2}{8r} \quad [\text{wb}] \text{ or } [\text{T} \cdot \text{m}^2] \quad (a < r < b)$$

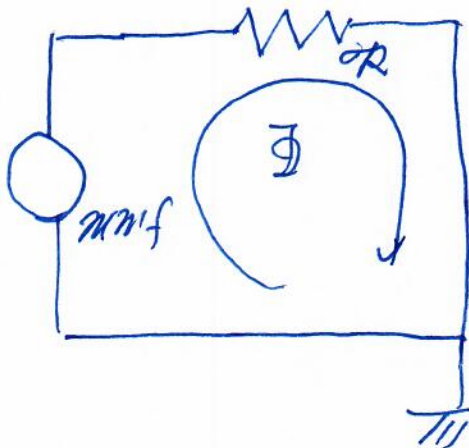
$$(d) \quad L = \frac{\Lambda}{I} = \frac{N\vec{\Phi}}{I} = \frac{N}{I} \left(\frac{N\mu_0 I (b-a)^2}{8r} \right) = \frac{N^2 \mu_0 (b-a)^2}{8r}$$

$$W_m = \int \vec{E} \cdot d\vec{v} = \frac{1}{2} LI^2 = \frac{1}{2} \left(\frac{N^2 \mu_0}{8r} \right) (b-a)^2 I^2$$

$$= \frac{N^2 \mu_0 (b-a)^2 I^2}{16r}$$

$$(e) \quad R = \frac{l_{\text{mean, path}}}{\mu_0 B}, \text{ where, } S = \pi \left(\frac{b-a}{2}\right)^2 \quad l_{\text{mean, path}} = 2\pi \left(\frac{b-a}{2}\right)$$

$$= \frac{2\pi \left(\frac{b-a}{2}\right)}{\mu_0 \pi \left(\frac{b-a}{2}\right)^2} = \frac{2}{\mu_0 \left(\frac{b-a}{2}\right)} = \frac{4}{\mu_0 (b-a)}$$

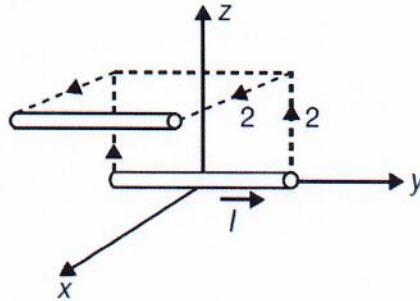


$$R = \frac{4}{\mu_0 (b-a)}$$

$$\text{MMF} = N\mu_0 I$$

$$\vec{\Phi} = \frac{N\mu_0 I (b-a)^2}{8r} \quad (a < r < b)$$

Q2. A conductor length 4 [m] , with current held at 10 [A] in the $\hat{a}_y$ direction, lies along the y axis between $y = \pm 2 \text{ [m]}$. If the $\vec{B} = 0.05\hat{a}_x \text{ [T]}$, find the force for the entire motion and the work in moving the conductor parallel to itself at constant speed to $x = z = 2 \text{ [m]}$. Refer to below Figure to understand the geometrical configuration.



At first, $\vec{F}$ due to $\vec{B}$ is needed to be determined,

$$\vec{F} = I(\vec{L} \times \vec{B}) \quad I = 10 \text{ [A]}, \quad \vec{L} = 4\hat{a}_y, \quad \vec{B} = 0.05\hat{a}_x$$

$$\vec{F} = 10(4\hat{a}_y \times 0.05\hat{a}_x) = \underline{2(-\hat{a}_z)} \quad \vec{F}_a = 2(\hat{a}_z)$$

Therefore, the work is moving the conductor to $x = z = 2$

$$W = \int_0^2 \vec{F}_a \cdot d\vec{l} + \int_0^2 \vec{F}_a \cdot d\vec{l} = \int_0^2 2\hat{a}_z \cdot \hat{a}_z dz + \int_0^2 \cancel{2\hat{a}_z \cdot \hat{a}_x dx} \quad \begin{matrix} \text{Due to} \\ \vec{F}_a = 2\hat{a}_z \end{matrix}$$

$$= 2 \int_0^2 dz = \underline{4 \text{ [J]}}$$

Q3. In a conductor wire with a radius of 0.5 [m] for which $\sigma = 5.0 \text{ [S/m]}$ and $\epsilon_r = 1$ the electric field intensity $\vec{E} = 250 \sin 10^{10} t \text{ [V/m]}$.

- Find the conduction and displacement current density.
- Find the frequency at which they have equal magnitudes.
- Determine the $\vec{H}$ at a distance b from the conducting wire.

Based on the time-harmonic Maxwell's Equation at the $\vec{E} = 250 \sin 10^{10} t$

$$\nabla \times \vec{H} = \vec{J}_c + \vec{J}_D = \sigma \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t}$$

$$(a) \quad \vec{J}_c = \sigma \vec{E} = \sigma (250 \sin 10^{10} t) = 5 (250 \sin 10^{10} t) = \underline{1250 \sin 10^{10} t \text{ [A/m}^2\text{]}}$$

$$\begin{aligned} \vec{J}_D &= \epsilon \frac{\partial \vec{E}}{\partial t} = \epsilon_0 \epsilon_r \frac{\partial}{\partial t} (250 \sin 10^{10} t) \\ &= \left(\frac{1}{36\pi} \times 10^{-9} \right) (1) (10^{10} \cdot 250 \cos 10^{10} t) = \underline{22.1 \cos 10^{10} t \text{ [A/m}^2\text{]}} \end{aligned}$$

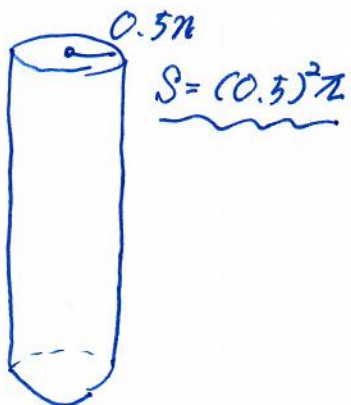
$$(b) \quad |\vec{J}_c| = |\vec{J}_D| \rightarrow |\sigma \vec{E}| = |j\omega \epsilon \vec{E}|, \text{ Therefore, } \sigma = \omega \epsilon$$

$$f = \frac{\sigma}{2\pi(\epsilon_0 \epsilon_r)} = \frac{5}{2\pi \left(\frac{1}{36\pi} \times 10^{-9} \right) (1)} \approx \underline{90 \text{ GHz}}$$

$$\begin{aligned} (c) \quad \oint_S (\vec{H}) \cdot d\vec{s} &= \oint_C \vec{H} \cdot d\vec{l} = \oint_C H_\phi \hat{\phi} \cdot \hat{\phi} b d\phi = 2\pi b H_\phi \\ &= \int_S (\vec{J}_c + \vec{J}_D) \cdot d\vec{s} = \vec{J}_c S + \vec{J}_D S \end{aligned}$$

where, $J_c = 1250 \sin 10^{10} t$, $J_D = 22.1 \cos 10^{10} t$

Therefore



$$2\pi b H_\phi = 0.25\pi [1250 \sin 10^{10} t + 22.1 \cos 10^{10} t]$$

$$\begin{aligned} H_\phi &= \frac{0.25\pi [1250 \sin 10^{10} t + 22.1 \cos 10^{10} t]}{2\pi b} \\ &= \underline{\underline{\frac{0.125 [1250 \sin 10^{10} t + 22.1 \cos 10^{10} t]}{b}}}} \end{aligned}$$

$$\vec{H} = \hat{\phi} \frac{0.125 [1250 \sin 10^{10} t + 22.1 \cos 10^{10} t]}{b}$$

Q4. An $\vec{H}$ field travels in the $-\hat{a}_z$ direction in free space with a **phase shift constant** (β) of 30.0 rad/m and an amplitude of $(1/3\pi) \text{ [A/m]}$. If the field has the direction $-\hat{a}_y$ when $t = 0$ and $z = 0$

- Write suitable expressions for $\vec{E}$ and $\vec{H}$.
- Determine the **angular frequency** and **wavelength**.
- Find the **wave number**.

To build up $\vec{H}$, [$-z$ travel direction, $H_m = (\frac{1}{3\pi})$, polarization direction, $(-\hat{a}_y)$]
 & for time-harmonic with $\omega = 2\pi f$.

$$\vec{H} = H_m \cos(\omega t - \beta(-z))(-\hat{a}_y) = (\frac{1}{3\pi}) \cos(\omega t + \beta z)(-\hat{a}_y)$$

$$\text{Where, } u = f \cdot \lambda = \frac{\omega}{\beta} \quad \beta = \frac{\omega}{u} = \frac{2\pi f}{f \cdot \lambda} = \frac{2\pi}{\lambda} = k$$

$$\beta = k (\text{Wave number}) = \text{phase shift constant} = \underline{30 \text{ rad/m}}$$

(a) $\vec{H} = (\frac{1}{3\pi}) \cos(\omega t + 30z)(-\hat{a}_y)$, $\vec{E}$, $\vec{H}$ are always travel to the same direction, so $\vec{E} \times \vec{H}$ has $(-z)$ -direction. Hence,

$\vec{E} \times \vec{H} \propto (-\hat{a}_z) \rightarrow \eta_0$ (Characteristic Impedance) in free space

$$\eta_0 = \frac{E_x}{H_y} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi, \text{ but } \vec{E} \times \vec{H} \text{ in } -z \text{ direction, so}$$

$$\frac{E_x}{H_y} = -\eta_0 = -120\pi. \quad \vec{E} = -\eta_0 \vec{H} = -120\pi \times (\frac{1}{3\pi}) \cos(\omega t + 30z) \hat{a}_x$$

$$\therefore \vec{E} = 40 \cos(\omega t + 30z) \hat{a}_x$$

$$(b) \quad \lambda = \frac{2\pi}{\beta} = \frac{2\pi}{30} = \frac{\pi}{15} \quad f = \frac{u}{\lambda} = \frac{3 \times 10^8 \text{ m/sec}}{\pi/15} = \frac{45}{\pi} \times 10^8 \text{ Hz}$$

$$(c) \quad k = \beta = \frac{\omega}{u} = \frac{\omega}{f \cdot \lambda} = \frac{2\pi f}{f \cdot \lambda} = \frac{2\pi}{\lambda} = \frac{2\pi}{\pi/15} = \underline{30 \text{ rad/m}}$$